Industrial Internship Report on

”E-commerce website for automative parts”

Prepared by

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Executive Summary
This project is a comprehensive automotive e-commerce web application developed using Flask, designed to facilitate a seamless shopping experience for users. The application allows users to register, log in, and manage their profiles, incorporating a custom CAPTCHA for enhanced security during the login process. Users can browse a wide range of automotive products categorized for easy navigation, add items to their shopping cart or wishlist, and proceed to checkout. The application supports order management, enabling users to view their order history and cancel orders if necessary. Upon successful order placement, users receive confirmation emails, enhancing communication and trust. The integration of SendGrid allows for efficient email notifications, including password recovery and acknowledgment of user queries submitted through a contact form. Admins have dedicated functionalities to manage the platform, including adding new products and categories, ensuring the inventory remains up-to-date. The application also supports file uploads, allowing users to upload profile images, which are securely stored. With a search feature for quick product discovery, this automotive e-commerce platform combines user-friendly design with robust functionality, making it an effective solution for online automotive shopping. Overall, it aims to provide a secure, efficient, and enjoyable shopping experience for all users.

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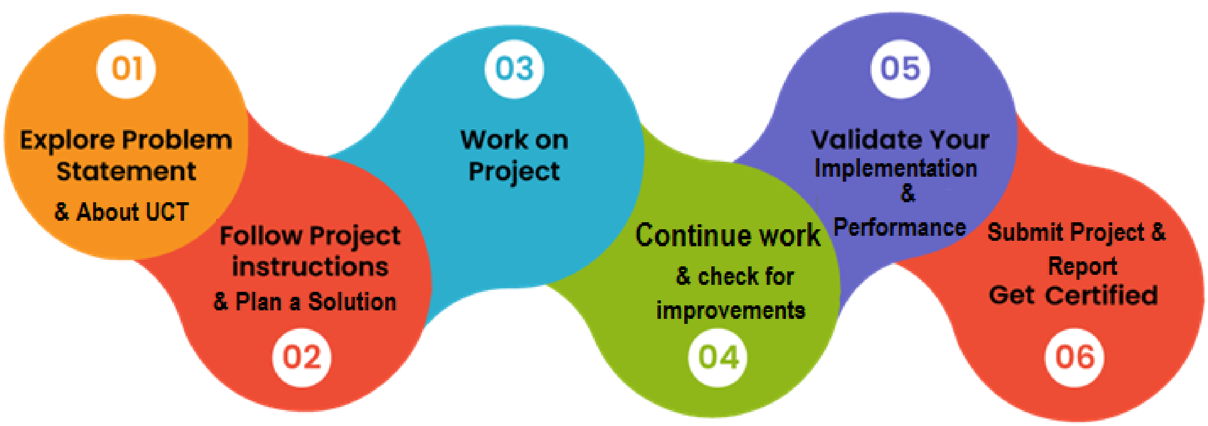
1 Preface

The automotive industry has witnessed a significant transformation with the advent of e-commerce, allowing consumers to purchase parts and accessories online with ease. This project aims to develop a comprehensive e-commerce platform specifically tailored for automotive parts, leveraging modern web technologies to enhance user experience and streamline the purchasing process. The platform is designed to cater to a diverse audience, including car enthusiasts, mechanics, and everyday consumers seeking reliable automotive components.

In this documentation, we will explore the various aspects of the project, from its inception to its implementation and testing. The project utilizes the Flask framework, SQLite3 database, HTML, and CSS, along with the SendGrid API for email functionalities. By integrating these technologies, we aim to create a robust and scalable application that meets the needs of users while ensuring security and performance.

The following sections will provide a detailed overview of the project, including its objectives, the problem it addresses, existing solutions, and the proposed design. We will also delve into the testing procedures employed to ensure the application's reliability and performance. Additionally, reflections on personal learnings and future work scope will be discussed, providing insights into the development process and potential enhancements.



This project is not just a technical endeavor; it represents a commitment to improving the online shopping experience for automotive parts. By focusing on user-centric design and functionality, we aim to bridge the gap between consumers and the automotive parts they need. The documentation serves as a comprehensive guide for stakeholders, developers, and users, offering a clear understanding of the project's goals, methodologies, and outcomes.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

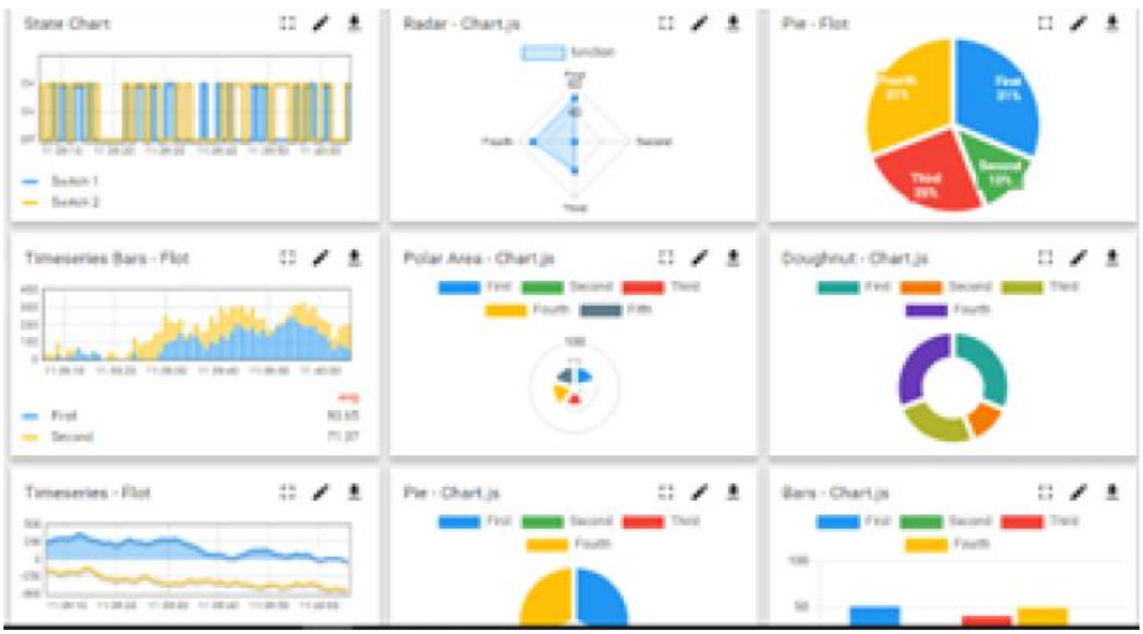
i. UCT IoT Platform (uct Insight)

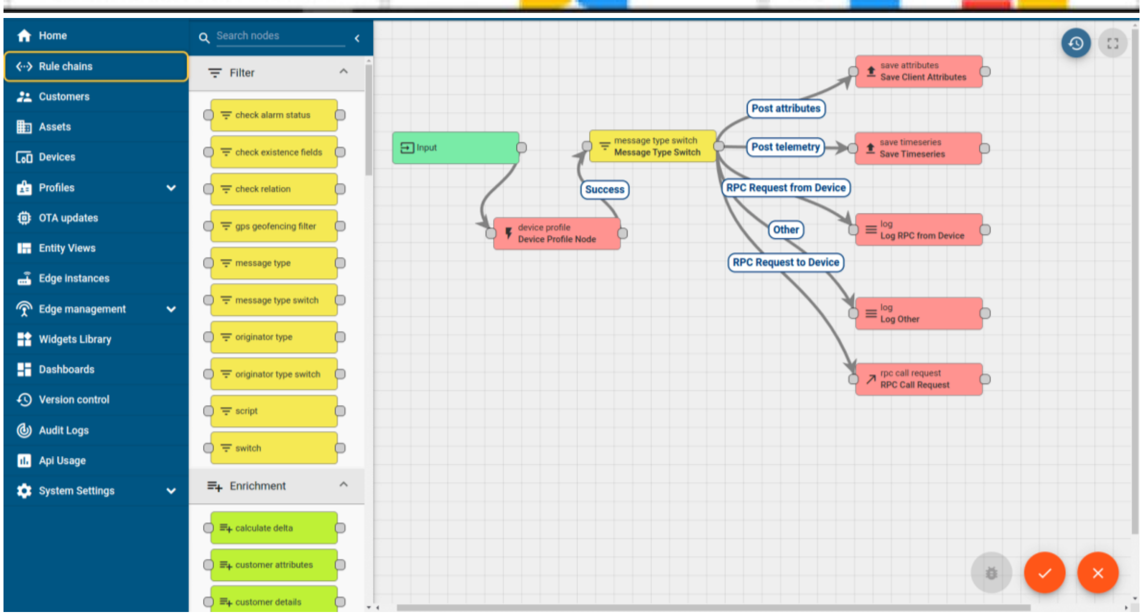
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

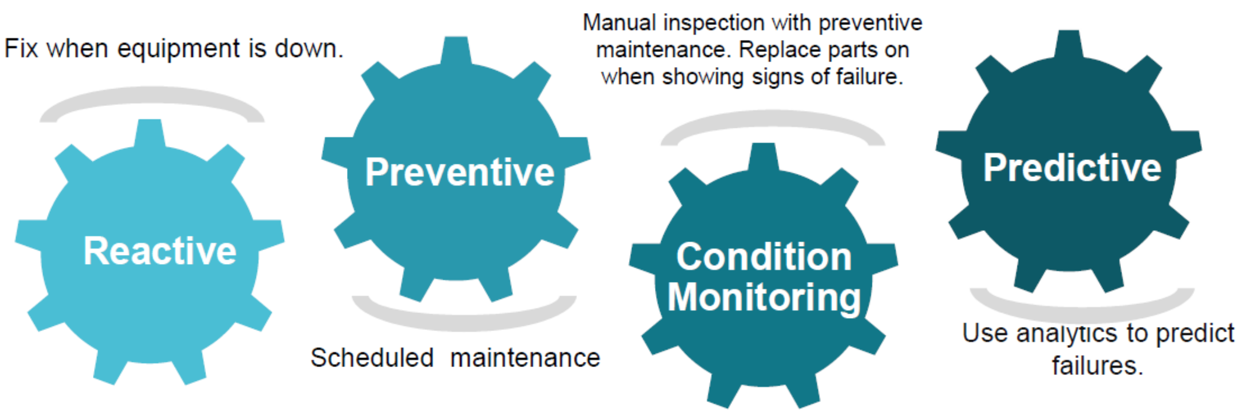
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

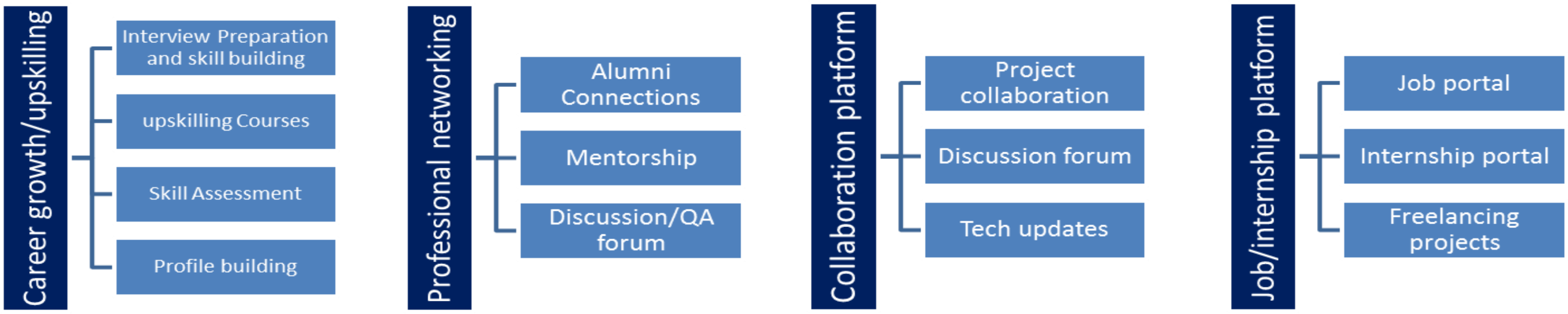
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] https://flask.palletsprojects.com/en/stable/

[2] https://flask-uploads.readthedocs.io/en/latest/

[3] https://www.twilio.com/docs/sendgrid

2.6 Glossary

Terms	Acronym
E-commerce	The buying and selling of goods or services over the internet, encompassing a wide range of online transactions.
F**lask**	A lightweight web framework for Python that allows developers to build web applications quickly and efficiently.
S**qlite**	A self-contained, serverless, and zero-configuration SQL database engine that is widely used for small to medium-sized applications.
API (Application Programming Interface):	A set of rules and protocols that allows different software applications to communicate with each other.
CRUD Operations	Refers to the four basic operations of persistent storage: Create, Read, Update, and Delete.

3 Problem Statement

The automotive parts market has traditionally faced several challenges that hinder the online shopping experience for consumers. Despite the growth of e-commerce, many existing platforms lack user-friendly interfaces, efficient search functionalities, and adequate customer support. This project aims to address these issues by developing a dedicated e-commerce platform for automotive parts that enhances user experience and simplifies the purchasing process.

One of the primary challenges consumers face is the difficulty in finding specific automotive parts online. Many existing platforms have poorly organized catalogs, making it hard for users to navigate and locate the products they need. Additionally, the search functionalities on these platforms are often inadequate, leading to frustration and abandoned shopping carts. This project seeks to implement a comprehensive product catalog with advanced search capabilities, allowing users to quickly and easily find the parts they require.

Another significant issue is the lack of personalized user experiences on many e-commerce sites. Consumers often have to create accounts to access certain features, but the registration processes can be cumbersome and time-consuming. This project aims to streamline user registration and authentication, providing a seamless onboarding experience that encourages users to engage with the platform.

Furthermore, existing platforms frequently struggle with order management and customer support. Users may encounter difficulties in tracking their orders, managing returns, or receiving timely assistance when issues arise. By implementing a robust order management system and integrating automated email notifications, this project will ensure that users are kept informed about their transactions and can easily manage their orders.

4 Existing and Proposed solution

The existing e-commerce landscape for automotive parts is characterized by several limitations that hinder user experience and satisfaction. Many platforms suffer from poor navigation, inadequate search functionalities, and a lack of personalized features. Users often find it challenging to locate specific parts due to disorganized catalogs and inefficient filtering options. Additionally, the registration and login processes on these platforms can be cumbersome, leading to high abandonment rates.

The proposed solution is to develop a dedicated e-commerce platform that addresses these shortcomings by providing a streamlined and user-friendly experience. The application will feature a well-organized product catalog that categorizes automotive parts effectively, allowing users to browse and search for items with ease. Advanced search functionalities will enable users to filter products based on various criteria, such as brand, price, and compatibility, ensuring they can quickly find the parts they need.

4.1 Code submission (Github link):

https://github.com/Sandeep2027/Upskill-Campus/tree/main/Full%20Stack%20Web%20Development

4.2 Report submission (Github link) :

https://github.com/Sandeep2027/Upskill-Campus/blob/main/Full%20Stack%20Web%20Development/SandeepPittala_EcommerceWebsiteForAutomativeParts.pdf

5 Proposed Design/ Model

The proposed design for the e-commerce platform is centered around a user-friendly interface and a robust architecture that supports scalability and performance. The application will be built using the Flask framework, which allows for rapid development and easy integration with various components. The architecture will follow a Model-View-Controller (MVC) pattern, separating the application logic, user interface, and data management to ensure maintainability and flexibility.

5.1 High Level Diagram (if applicable)

The high-level diagram of the e-commerce platform illustrates the overall architecture and the interactions between various components. At the core of the architecture is the Flask web application, which serves as the central hub for handling user requests and managing data. The application interacts with the SQLite3 database to store and retrieve information related to users, products, orders, and categories.

The front-end of the application communicates with the back-end through RESTful APIs, allowing for efficient data exchange. Users access the platform via web browsers, where they can browse products, manage their accounts, and complete transactions. The diagram also highlights the integration of the SendGrid API for email notifications, ensuring that users receive timely updates regarding their orders and account activities.

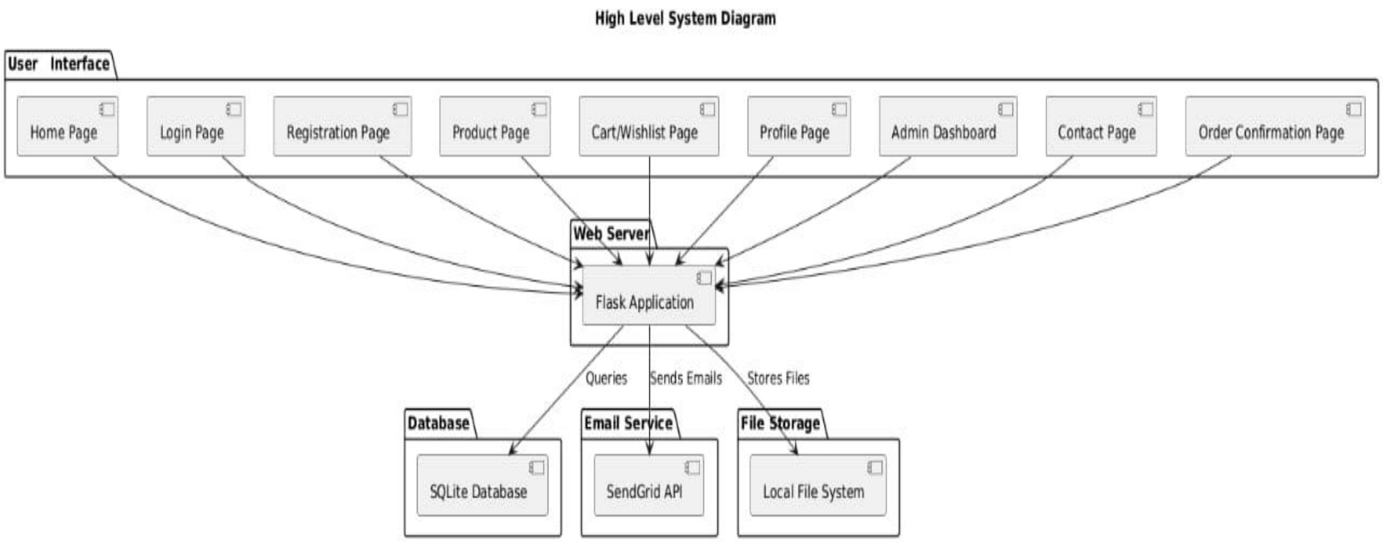


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

The low-level diagram provides a more detailed view of specific components within the e-commerce platform, focusing on the interactions between the database, application logic, and user interface. It outlines the various entities within the SQLite3 database, such as Users, Products, Orders, and Categories, along with their attributes and relationships.

For instance, the Users table may include fields such as user_id, email, password_hash, and profile_info, while the Products table could contain product_id, name, description, price, and stock_quantity. The Orders table would link users to their purchases, including order_id, user_id, product_id, order_date, and status. This detailed representation helps in understanding how data is structured and accessed within the application.

5.3 Interfaces (if applicable)

The interfaces of the e-commerce platform encompass both user interfaces and application programming interfaces (APIs) that facilitate communication between different components. The user interface is designed to be intuitive and responsive, ensuring that users can easily navigate the platform and access the features they need.

Key user interfaces include the Home Page, Product Pages, Login and Registration Forms, Shopping Cart, and Admin Dashboard. Each interface is designed with user experience in mind, incorporating elements such as search bars, filters, and clear calls to action. The design ensures that users can quickly find products, manage their accounts, and complete transactions without unnecessary friction.

6 Performance Test

Performance testing is a critical aspect of the development process, ensuring that the e-commerce platform can handle user load effectively and deliver a seamless experience. This section outlines the testing strategy employed to evaluate the application's performance, including the test plan, procedures, and outcomes.

6.1 Test Plan/ Test Cases

The performance testing plan includes various test cases to evaluate the application's response times, throughput, and resource utilization:

1. **Load Testing**: Simulates multiple users accessing the platform simultaneously to assess performance under heavy load, measuring response times and identifying bottlenecks.
2. **Stress Testing**: Tests the application beyond its limits to find its breaking point and evaluate recovery capabilities under extreme conditions.
3. **Endurance Testing**: Runs the application under sustained load for an extended period to check stability and identify memory leaks or performance degradation.
4. **Spike Testing**: Introduces sudden traffic spikes to assess the application's ability to handle abrupt changes in load and scale effectively.
5. **Scalability Testing**: Evaluates the application's capacity to scale horizontally or vertically to meet increased user demand.

6.2 Test Procedure

1. The performance testing procedure involves:
2. Setting up a testing environment that closely mirrors production.

3. Using tools like JMeter or LoadRunner to simulate user interactions through automated test scripts.
4. Executing tests to generate load and collect performance metrics.
5. Analyzing results to identify performance issues and areas for improvement.

6.3 Performance Outcome

The outcomes of the performance tests provide insights into the application's behavior:

- **Load Testing Results**: The application managed a specified number of concurrent users with acceptable response times, indicating efficient handling of user requests.
- **Stress Testing Results**: Identified the breaking point where performance declined, highlighting areas needing optimization for better resilience.
- **Endurance Testing Results**: Showed stability under sustained load, but minor memory leaks were detected, indicating a need for further optimization.
- **Spike Testing Results**: The application handled sudden traffic spikes well, though minor delays were noted, suggesting additional resources may be needed.
- **Scalability Testing Results**: Confirmed the application can scale effectively to support increased user demand, ensuring responsiveness as the user base grows.

7 My learnings

During the development of the e-commerce platform for automotive parts, I gained valuable insights that contributed to my growth. Key learnings include:

1. **Technical Skills**: Enhanced my web development skills using the Flask framework and the MVC pattern, along with hands-on experience in SQLite3 for database design and optimization.
2. **Front-End Development**: Designed user interfaces with HTML, CSS, and JavaScript, focusing on responsive design and user-centric principles.
3. **API Integration**: Learned to integrate the SendGrid API for email notifications, understanding how APIs facilitate communication within applications.
4. **Project Management**: Gained experience in agile methodologies, emphasizing iterative development, task prioritization, and incorporating user feedback.
5. **Problem-Solving**: Developed strong problem-solving and debugging skills by addressing various challenges throughout the project.

6. **Collaboration and Communication**: Emphasized the importance of clear communication and teamwork, learning to articulate ideas and provide constructive feedback.
7. **User Experience Focus**: Reinforced the significance of understanding user needs to guide design decisions and meet audience expectations.

8 Future work scope

The e-commerce platform for automotive parts is set for future enhancements, focusing on improving user experience and functionality. Key areas for development include:

1. **Mobile App Development**: Creating a mobile app to enhance accessibility and user engagement.
2. **Advanced Search and Filtering**: Implementing sophisticated search features like voice search and personalized recommendations.
3. **User Reviews and Ratings**: Adding a review system to build community trust and assist in product decisions.
4. **Enhanced Admin Dashboard**: Expanding analytics tools for better insights into user behavior and sales trends.
5. **Third-Party Integration**: Partnering with logistics services for improved order processing and shipping.
6. **Loyalty Program**: Introducing a rewards system to encourage repeat purchases and customer retention.
7. **Internationalization**: Supporting multiple languages and currencies to reach a global audience.
8. **AI Chatbot for Support**: Integrating a chatbot for instant customer assistance.
9. **Performance Optimization**: Regularly testing and optimizing the platform to maintain efficiency.
10. **User Feedback Mechanism**: Establishing a system for users to share suggestions to guide future improvements.